

Michael Beyeler



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Email: mbeyeler@ucsb.edu
Lab: bionicvisionlab.org
Faculty Profile: CS, PBS

ACADEMIC APPOINTMENTS

- **Associate Professor** · Computer Science (CS) · Psychological & Brain Sciences (PBS) 2025 – present
Associate Director · Research Center for Virtual Environments and Behavior (ReCVEB)
University of California, Santa Barbara (UCSB)
Affiliations: Electrical & Computer Engineering (ECE) · Dynamical Neuroscience (DYNS)
- **Assistant Professor** · Computer Science (CS) · Psychological & Brain Sciences (PBS) 2019 – 2025
University of California, Santa Barbara (UCSB)
- **Postdoctoral Fellow** · Psychology · Institute for Neuroengineering · eScience Institute 2016 – 2019
University of Washington (UW)

EDUCATION

- **PhD in Computer Science** · Specialization in Computational Neuroscience 2012 – 2016
University of California, Irvine (UCI)
Dissertation: Cortical neural network models of visual motion perception for decision-making and reactive navigation, May 2016. Committee: JL Krichmar (co-chair), N Dutt (co-chair), C Fowlkes
- **MS in Biomedical Engineering** · Focus on Bioelectronics 2009 – 2011
ETH Zurich, Switzerland
- **BS in Electrical Engineering** · Major in Micro- and Optoelectronics 2005 – 2009
ETH Zurich, Switzerland

HONORS & AWARDS

Major Fellowships, Honors & Awards

- Harold J. Plous Memorial Award: *UCSB* 2024
- DP2 New Innovator Award: *National Institutes of Health (NIH)* 2022
- K99/R00 Pathway to Independence Award: *NIH* 2018
- Innovation in Neuroengineering & Data Science Postdoctoral Fellowship: *Gordon & Betty Moore Foundation, Alfred P. Sloan Foundation, Washington Research Foundation (WRF)* 2016

Best Paper Award Nominations

- Honorable Mention Best Paper Award (top 4%): C9, *Augmented Humans (AHs)* 2021
- Nominee: Best Student Paper, C6, *IEEE International Joint Conference on Neural Networks (IJCNN)* 2018
- Nominee: Best Student Paper, C1, *IEEE Biomedical Circuits & Systems Conference (BioCAS)* 2010

Other Conference Awards

- Best Poster Award (best of 15): W3, *Augmented Humans (AHs)* 2022
- Abstract of Distinction (top 3%): A34, *Association for Research in Vision & Ophthalmology (ARVO)* 2020
- Best Poster Award: A19, *Eye & Chip World Congress on Artificial Vision* 2017
- Best Workshop Talk Award: A6, *IEEE International Conference on Robotics & Automation (ICRA)* 2014

Other Academic Award Nominations

- Nominee: Academic Senate Outstanding Graduate Mentor Award, *UCSB* 2022, 2025

RESEARCH GRANTS & OTHER SUPPORT*Our share, total: \$4.1M, as sole PI: \$3.0M***Active Funding**

- EEG signatures of TMS-induced visual perception in sighted and blind individuals 2025 – present
M Beyeler, R Lapate, TC Sprague (Our share: \$10,988)
- Attentional dynamics during navigation: Strategies, individual differences, and neural mechanisms, *Institute for Collaborative Biotechnologies (ICB)*. MPI: B. Giesbrecht, **M. Beyeler**, M. Hegarty. (Our share: \$150,000) 2024 – present
- DP2 LM014268: Towards a *Smart Bionic Eye*: AI-powered artificial vision for the treatment of incurable blindness, *NIH*. PI: **M Beyeler**. (\$1,991,522) 2022 – present

Completed Funding

- Computational virtual patient to predict perceptual capabilities of prosthetic vision, *UC Noyce Initiative*. PI: MP Eckstein, Co-PI: **M Beyeler**. (Our share: \$100,000) 2023 – 2025
- R01 NS121919: Cortical visual processing for navigation, *NIH*. 2021 – 2026
PI: S Smith. MPIs: M Goard, C Niell. Co-I: **M Beyeler**. (Our share: \$718,387)
- Visual navigation under high-stress conditions: Improving situational awareness through deep-learning based vision augmentation in immersive virtual training environments, *ICB*. MPI: **M Beyeler**, M Hegarty, S Grafton, B Giesbrecht. (Our share: \$150,000) 2021 – 2024
- K99/R00 EY029329: Virtual prototyping for retinal prosthesis patients, *NIH*. 2018 – 2023
PI: **M Beyeler**. (\$968,319)
- Event-based scene understanding for bionic vision, *UCSB Academic Senate Research Faculty Grant*. PI: **M Beyeler**. (\$10,000) 2021 – 2022
- An inaugural data science summit at UCSB, *Academic Data Science Alliance (ADSA)* 2021
PI: A Frank. Co-PIs: A Horst, **M Beyeler**. (\$9,258)
- Eye tracking in immersive virtual environments, *UCSB Academic Senate Research Faculty Grant*. PI: M Hegarty. Co-PI: **M Beyeler**. (\$5,099) 2020 – 2021

ACADEMIC MENTORING**Postdoctoral Scholars***Total: 5*

- Marius Schneider, ICB 2024 – present
- Jacob Granley, CS (Next position: ML Engineer at Neuralink) 2025 – 2026
- Roksana Sadeghi, CS (Next position: Health Studies Scientist at Apple) 2025
- Amirali Vahid, ICB (Next position: Postdoc at Stanford Medicine) 2022 – 2023
- Melani Sanchez Garcia, CS (Next position: Senior AI Researcher at Kimera Technologies) 2022 – 2023

PhD Advisees · Chair*Total: 11*

- Emily M. Joyce, Dynamical Neuroscience (DYNS) 2025 – present
- Hannah L. Stone, PBS 2024 – present
- Lucas Gil Nadolskis, DYNS 2023 – present
- Apurv Varshney, CS 2023 – present
- Lily M. Turkstra, PBS 2023 – present
- Galen Pogoncheff, CS 2022 – present
- Yuchen Hou, CS 2022 – present
- Byron A. Johnson, PBS (Co-chair: Miguel Eckstein; Next position: Postdoc at UC Berkeley) 2020 – 2025
- Jacob Granley, CS (Next position: Postdoc in the Bionic Vision Lab) 2020 – 2025
- Aiwen Xu, CS (Next position: Software Engineer at Snowflake AI) 2020 – 2024
- Justin M. Kasowski, DYNS (Next position: Founder & CEO at RealmVR) 2019 – 2023

PhD Advisees · Committee Member

Total: 13

PhD Advisees · External Examiner

Total: 5

- Vaishnavi Mohan, *University of Washington* W'25 – present
- Jack White, *Swinburne University of Technology, Melbourne, Australia* W'22
- Melani Sanchez Garcia, *Universad de Zaragoza, Spain* W'22
- Tristan Fauvel, *Institute de la Vision, Sorbonne Université, Paris, France* F'21
- Kexin Chen, *Cognitive Sciences, UCI* S'20

MS Advisees · Chair

Total: 8

MS Advisees · Committee Member

Total: 6

Lab Managers

Total: 5

UC LEADS Mentorship Program Advisees

Total: 2

Undergraduate Honor Advisees

Total: 10

Undergraduate Research Assistants

Total: 84+

High School Advisees

Total: 11

ACADEMIC SERVICE**University-Wide Committees**

- Member, CS Representative: Faculty Legislature, *UCSB* 2020 – 2022
- Postdoctoral Representative: Research Advisory Board, *UW* 2017 – 2019

Departmental Committees

- Member: Graduate Affairs Committee, *CS, UCSB* 2025 – present
- Member: Shared Lab Coordinators, *PBS, UCSB* 2025 – present
- Chair: Public Relations & Awards Committee, *CS, UCSB* 2023 – present
- Member: Graduate Admissions Committee, *CS, UCSB* 2023 – 2024
- Member: Diversity, Equity, and Inclusion Committee, *CS, UCSB* 2022 – 2025
- Member: Graduate Admissions Committee, *DYNS, UCSB* 2021 – 2023
- Member: Graduate Admissions Committee, *CS, UCSB* 2019 – 2020

Organized Workshops, Competitions & Summits

- Organization Committee Member: Mouse vs. AI: A Neuroethological Benchmark for Visual Robustness, *NeurIPS* 2025
- Applications Chair-Elect: Optica Fall Vision Meeting 2024 – present
- Steering Committee Member: 2022 Mind & Machine Intelligence Summit, *UCSB* 2021 – 2022
- Co-organizer: 2021 UCSB Data Science Summit, *UCSB* 2020 – 2021
- Organizer: Recent Computational Advances in Neuroengineering, Workshop, *Computational & Systems Neuroscience (COSYNE)* 2018

Editorial Boards

- Review Editor: *Frontiers in Human Neuroscience* 2020 – 2025
- Guest Editor: *eLife* 2023
- Review Editor: *Frontiers in Neurorobotics* 2017 – 2020

Ad-Hoc Reviewing · Grants

- Ad-hoc reviewer, ZRG1 IBV-P (01), *Imaging & Bioengineering Technology for Visual Systems* 2025

- Ad-hoc reviewer, ZEB1 OSR-H (O1) R, *Bioethics & Tech Development*, *NIH* 2024
- Ad-hoc reviewer, *Army Research Office (ARO)* 2024
- Ad-hoc reviewer, ZRG1 NV-P (81) S, *BIVT*, *NIH* 2023, 2024
- Ad-hoc reviewer, CDMRP VRP TECH, *Department of Defense (DoD)* 2022
- Ad-hoc reviewer, FOReSIGHT, *Institut Hospitalo-Universitaire (IHU)*, *Paris, France* 2022
- Ad-hoc reviewer, ZGM1 RCB-9 (CG), *BIVT*, *NIH* 2021
- Early Career Reviewer (ECR), ZRG1 ETTN-P (81), *NIH* 2021

Ad-Hoc Reviewing · Selected Journals

webofscience.com/wos/author/record/AFK-1674-2022

Communications Biology · eLife · Frontiers in Human Neuroscience · Frontiers in Neuroscience · IEEE Transactions on Neural Networks & Learning Systems (TNNLS) · Journal of Neural Engineering · Journal of Neuroscience · Journal of Vision · Journal of Visual Impairment & Blindness (JVIB) · Nature · Nature Biomedical Engineering · Neural Networks · PLoS Computational Biology · Science Advances · Vision Research

Ad-Hoc Reviewing · Selected Conferences

ACM Conference on Human Factors in Computing Systems (CHI) · Computational & Systems Neuroscience (COSYNE) · IEEE Conference on Virtual Reality and 3D User Interfaces (VR) · IEEE International Conference on Intelligent Robots & Systems (IROS) · IEEE International Symposium on Circuits & Systems (ISCAS) · IEEE International Symposium on Mixed and Augmented Reality (ISMAR) · Medical Image Computing & Computer Assisted Intervention (MICCAI) · Scientific Computing with Python (SciPy)

PUBLICATIONS

scholar.google.com/citations?user=dK-0kG4AAAAJ

Note that in many areas of computer science, *conferences* are the primary venue for peer-reviewed publications, with selectivity and impact often exceeding that of journals (Chen & Konstan, 2010). The opposite is true in neuroscience. Legend: */** equal contribution, [Ⓜ] invited publication, [Ⓡ] review/survey article

Refereed Journal Articles

- J27 LM Turkstra*, BA Johnson*, A Kartha, G Dagnelie, **M Beyeler** (2026). Gamification enhances user engagement and task performance in prosthetic vision testing. *Translational Vision Science & Technology*
- J26 BA Johnson, AK Soni, S Murlidaran, **M Beyeler**, MP Eckstein (2026). Distinct roles of central and peripheral vision in rapid scene understanding. *Journal of Vision*
- J25 CM Niell*, **M Beyeler***, MJ Goard*, SL Smith* (2026). Ecological visual processing in the mouse. *Annual Reviews of Neuroscience*
- J24 M Downing, M Peng, J Granley, **M Beyeler**, T Bultan (2026). Fuzzing the brain: Automated stress testing for the safety of ML-driven neurostimulation. *Journal of Neural Engineering* 23 026004.
- J23 RB Esquenazi, K Meier, **M Beyeler**, D Wright, GM Boynton, I Fine (2025). Perceptual learning of prosthetic vision using video game training. *Journal of Vision* 25(12):12.
- J22 JM Kasowski, A Varshney, R Sadeghi, **M Beyeler** (2025). Simulated prosthetic vision confirms checkerboard as an effective raster pattern for epiretinal implants. *Journal of Neural Engineering* 22 046017.
- J21 LM Turkstra, T Bhatia, A Van Os, **M Beyeler** (2025). Assistive technology use in domestic activities by people who are blind. *Scientific Reports* 15:7486.
- J20 L Nadolskis*, LM Turkstra*, E Larnyo, **M Beyeler** (2024). Aligning visual prosthetic development with implantee needs. *Translational Vision Science & Technology* Vol.13, 28.
- J19 Y Hou, D Nanduri, JD Weiland, **M Beyeler** (2024). Axonal stimulation affects the linear summation of single-point perception in three Argus II users. *Journal of Neural Engineering* 21 026031 [Code] [Data]
- J18 G Pogoncheff, Z Hu, A Rokem, **M Beyeler** (2024). Explainable machine learning predictions of perceptual sensitivity for retinal prostheses. *Journal of Neural Engineering* 21 026009 [Code]
- J17 A Varshney*, M Munns*, J Kasowski, M Zhou, C He, S Grafton, B Giesbrecht, M Hegarty, **M Beyeler** (2024). Stress affects navigation strategies in immersive virtual reality. *Scientific Reports*

- J16 A Xu, **M Beyeler** (2023). Retinal ganglion cells undergo cell type-specific functional changes in a computational model of cone-mediated retinal degeneration. *Frontiers in Neuroscience* 17:1147729 [Code]
- J15 J Kasowski*, BA Johnson*, R Neydavood, A Akkaraju, **M Beyeler** (2023). A systematic review of extended reality (XR) for understanding and augmenting vision loss®. *Journal of Vision* 23(5):5, 124 **Cover article**
- J14 M Sanchez-Garcia*, T Chauhan*, BR Cottureau**, **M Beyeler**** (2023). Efficient multi-scale representation of visual objects using a biologically plausible spike-latency code and winner-take-all inhibition. *Biological Cybernetics* 117:95111
- J13 **M Beyeler**, M Sanchez-Garcia (2022). Towards a *Smart Bionic Eye*: AI-powered artificial vision for the treatment of incurable blindness. *Journal of Neural Engineering* 19:063001.
- J12 El Yücel, R Sadeghi, A Kartha, SR Montezuma, G Dagnelie, A Rokem, GM Boynton, I Fine, **M Beyeler** (2022). Factors affecting two-point discrimination in Argus II patients. *Frontiers in Neuroscience* 16:901337.
- J11 K Chen, **M Beyeler**, JL Krichmar (2022). Cortical motion perception emerges from dimensionality reduction with evolved spike-timing dependent plasticity rules. *Journal of Neuroscience*. **Featured research article**
- J10 RB Esquenazi, KM Meier, **M Beyeler**, GM Boynton, I Fine (2021). Learning to see again: Perceptual learning of simulated abnormal on- off- cell population responses in sighted individuals. *Journal of Vision* 21(13): 1–20.
- J9 BW Brunton, **M Beyeler** (2019). Data-driven models in human neuroscience and neuroengineering[®]. *Current Opinion in Neurobiology* 58: 21–29.
- J8 **M Beyeler**, D Nanduri, JD Weiland, A Rokem, GM Boynton, I Fine (2019). A model of ganglion axon pathways accounts for percepts elicited by retinal implants. *Scientific Reports* 9(1):9199. [Code] [Data]
- J7 **M Beyeler** (2019). Commentary: Detailed visual cortical responses generated by retinal sheet transplants in rats with severe retinal degeneration. *Frontiers in Neuroscience* 13: 471.
- J6 **M Beyeler***, EL Rounds*, KD Carlson, N Dutt, JL Krichmar (2019). Neural correlates of sparse coding and dimensionality reduction®. *PLOS Computational Biology* 15(6):e1006908.
- J5 **M Beyeler**, A Rokem, GM Boynton, I Fine (2017). Learning to see again: Biological constraints on cortical plasticity and the implications for sight restoration technologies®. *Journal of Neural Engineering* 14(5). **Featured cover article**
- J4 **M Beyeler**, N Dutt, JL Krichmar (2016). 3D visual response properties of MSTd emerge from an efficient, sparse population code. *Journal of Neuroscience* 36(32): 8399–8415.
- J3 **M Beyeler**, N Oros, N Dutt, JL Krichmar (2015). A GPU-accelerated cortical neural network model for visually guided robot navigation. *Neural Networks* 72: 75–87.
- J2 **M Beyeler**, M Richert, ND Dutt, JL Krichmar (2014). Efficient spiking neural network model of pattern motion selectivity in visual cortex. *Neuroinformatics*, 1–20.
- J1 **M Beyeler**, ND Dutt, JL Krichmar (2013). Categorization and decision-making in a neurobiologically plausible spiking network using a STDP-like learning rule. *Neural Networks* 48C: 109–124.

Refereed Conference Publications

- C27 G Pogoncheff, A Wang, J Granley, **M Beyeler** (2026). Percept-aware surgical planning for visual cortical prostheses with vascular avoidance. *Medical Image Computing & Computer Assisted Intervention (MICCAI)*, Strasbourg, France.
- C26 J Liu, Y Wang, **M Beyeler** (2026). Network-adaptive cloud preprocessing for visual neuroprostheses. *48th International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, Toronto, Canada.
- C25 J Lesner, **M Beyeler** (2026). SymbolSight: Minimizing inter-symbol interference for reading with prosthetic vision. *48th EMBC*, Toronto, Canada.
- C24 G Pogoncheff, **M Beyeler** (2026). BIRD: Behavior induction via representation-structure distillation. *14th International Conference on Learning Representations (ICLR 26)*, Rio de Janeiro, Brazil.
- C23 JM Kasowski, A Varshney, **M Beyeler** (2025). Static or temporal? Semantic scene simplification to aid wayfinding in immersive simulations of bionic vision. *31st ACM Symposium on Virtual Reality Software and Technology (VRST) '25*, Montreal, Canada.

- C22 R Sadeghi, **M Beyeler** (2025). Efficient spatial estimation of perceptual thresholds for retinal implants via Gaussian process regression. *47th EMBC*, Copenhagen, Denmark.
- C21 E Schoinas, A Rastogi, A Carter, J Granley, **M Beyeler** (2025). Evaluating deep human-in-the-loop optimization for retinal implants using sighted participants. *47th EMBC*, Copenhagen, Denmark.
- C20 R Gretschi, **M Beyeler**, J Lau, T Sherwood (2025). Single Spike Artificial Neural Networks. *Proceedings of the 52nd Annual International Symposium on Computer Architecture (ISCA)*, Tokyo, Japan.
- C19 Y Hou*, L Püllela*, J Su, S Aluru, X Lu, **M Beyeler** (2024). Predicting the temporal dynamics of prosthetic vision. *46th EMBC*, Orlando, FL. (oral) [Code]
- C18 A Xu, Y Hou, CM Niell, **M Beyeler** (2023). Multimodal deep learning model unveils behavioral dynamics of V1 activity in freely moving mice. *37th Conference on Neural Information Processing Systems (NeurIPS)*, New Orleans, LA. [Code] [Data]
- C17 J Granley, T Fauvel, M Chalk, **M Beyeler** (2023). Human-in-the-loop optimization for deep stimulus encoding in visual prostheses. *37th NeurIPS*, New Orleans, LA. [Code]
- C16 G Pogoncheff, J Granley, **M Beyeler** (2023). Explaining V1 properties with a biologically constrained deep learning architecture. *37th NeurIPS*, New Orleans, LA. [Code]
- C15 A Rasla, **M Beyeler** (2022). The relative importance of depth cues and semantic edges for indoor mobility using simulated prosthetic vision in immersive virtual reality. *ACM Symposium on Virtual Reality Software and Technology (VRST)*, Virtual/Tsukuba, Japan.
- C14 J Granley, L Relic, **M Beyeler** (2022). Hybrid neural autoencoders for stimulus encoding in visual and other sensory neuroprostheses. *36th NeurIPS*, New Orleans, LA.
- C13 A Bruce, **M Beyeler** (2022). Greedy optimization of electrode arrangement for epiretinal prostheses. *Medical Image Computing & Computer Assisted Intervention (MICCAI)*, Singapore. **Highlighted in MICCAI Daily**
- C12 J Kasowski, **M Beyeler** (2022). Immersive virtual reality simulations of bionic vision. *Augmented Humans (AHs)*, online. [Code] [Video]
- C11 J Granley, **M Beyeler** (2021). A computational model of phosphene appearance for epiretinal prostheses. *International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, online. [Code]
- C10 Z Hu, **M Beyeler** (2021). Explainable AI for retinal prostheses: Predicting electrode deactivation from routine clinical measures. *IEEE EMBS Conference on Neural Engineering (NER)*, online.
- C9 N Han, S Srivastava*, A Xu*, D Klein, **M Beyeler** (2021). Deep learning-based scene simplification for bionic vision. *Augmented Humans (AHs)*, online. [Code] [Data] **Honorable Mention Award (top 4 %)**
- C8 **M Beyeler**, GM Boynton, I Fine, A Rokem (2019). Model-based recommendations for optimal surgical placement of epiretinal implants. *Medical Image Computing & Computer Assisted Intervention (MICCAI)*, Shenzhen, China.
- C7 **M Beyeler** (2019). Biophysical model of axonal stimulation in epiretinal visual prostheses. *IEEE EMBS Conference on Neural Engineering (NER)*, San Francisco, CA.
- C6 T-S Chou*, HJ Kashyap*, J Xing, S Listopad, EL Rounds, **M Beyeler**, N Dutt, JL Krichmar (2018). CARLsim 4: An open source library for large scale, biologically detailed spiking neural network simulations using heterogeneous clusters. *IEEE International Joint Conference on Neural Networks (IJCNN)*, Rio de Janeiro, Brazil. **Best Student Paper Nominee**. [Code]
- C5 **M Beyeler**, GM Boynton, I Fine, A Rokem (2017). pulse2percept: A Python-based simulation framework for bionic vision. *Scientific Computing with Python (SciPy)*, p.81–88. [Code] [Talk]
- C4 **M Beyeler***, KD Carlson*, T-S Chou*, N Dutt, JL Krichmar (2015). CARLsim 3: A user-friendly and highly optimized library for the creation of neurobiologically detailed spiking neural networks. *IEEE International Joint Conference on Neural Networks (IJCNN)*, Killarney, Ireland. [Code]
- C3 KD Carlson, **M Beyeler**, N Dutt, JL Krichmar (2014). GPGPU accelerated simulation & parameter tuning for neuromorphic applications[Ⓢ]. *Asia and South Pacific Design Automation Conference (ASP-DAC)*, Singapore.
- C2 **M Beyeler**, F Mirus, A Verl (2014). Vision-based robust road lane detection in urban environments. *IEEE International Conference on Robotics & Automation (ICRA)*, Hong Kong, China.

- C1 **M Beyeler***, F Stefanini*, H Proske, CG Galizia, E Chicca (2010). Exploring olfactory sensory networks: simulations and hardware emulation. *IEEE Biomedical Circuits & Systems Conference (BioCAS)*, Paphos, Cyprus. **Best Student Paper Nominee.**

Refereed Workshop & Lightly Reviewed Short Papers

- W10 **M Beyeler** (2025). Bionic vision as neuroadaptive XR: Closed-loop perceptual interfaces for neurotechnology. *ISMAR-Adjunct*, Daejeon, South Korea
- W9 A Arora, LG Nadolskis, **M Beyeler**, M Sra (2024). Vision AI - Shopping assistance for people with vision impairments. *ISMAR-Adjunct*, Seattle, WA
- W8 J Granley*, G Pogoncheff*, A Rodil, L Soo, LM Turkstra, LG Nadolskis, A Alfaro Saez, C Soto Sanchez, E Fernandez Jover, **M Beyeler** (2024). Beyond sight: Probing alignment between image models and blind V1. *Workshop on Representational Alignment (Re-Align)*, ICLR '24, Vienna, Australia. **Spotlight Talk**
- W7 S Awasthi, V Ross, S Lim, **M Beyeler**, T Höllerer (2024). Eye tracking performance in mobile mixed reality. *IEEE VR*, Orlando, FL
- W6 S Awasthi, V Ross, **M Beyeler**, T Höllerer (2023). Eye Tracking Test Suite: Evaluating and calibrating eye tracking for mixed-reality locomotion. *ISMAR-Adjunct*, Sydney, Australia
- W5 J Granley, A Riedel, **M Beyeler** (2022). Adapting brain-like neural networks for modeling cortical visual prostheses. *SVRHM Workshop, NeurIPS*, New Orleans, LA.
- W4 M Sanchez-Garcia, T Chauhan, BR Cottureau, **M Beyeler** (2022). Efficient visual object representation using a biologically plausible spike-latency code and winner-take-all inhibition. *NeuroVision Workshop, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, online.
- W3 L Relic, B Zhang, YL Tuan, **M Beyeler** (2022). Deep learning-based perceptual stimulus encoder for bionic vision. *Augmented Humans (AHs)*, online. [Video] **Best Poster Award**
- W2 S Tang, Z Qi, J Granley, **M Beyeler** (2021). U-Net with hierarchical bottleneck attention for landmark detection in fundus images of the degenerated retina. *OMIA8 Workshop, MICCAI*, online.
- W1 J Kasowski, N Wu, **M Beyeler** (2021). Towards immersive virtual reality simulations of bionic vision. *Augmented Humans (AHs)*, online.

US Patents

- P1 R Appuswamy, **M Beyeler**, P Datta, MD Flickner, DS Modha (2023). Long Short-Term Memory (LSTM) cells on spiking neuromorphic hardware. US Patent No. 11,636,317.

Selected Contributed Abstracts & Poster Presentations

- A62 LM Turkstra, BA Johnson, A Kartha, G Dagnelie, **M Beyeler** (2026). Perceptual Integration Window in Argus II: Millisecond Timing and Axon-Bundle Geometry Determine Two-Point Perception. *Association for Research in Vision & Ophthalmology (ARVO)* '26, Denver, CO.
- A59 EM Joyce, N Thomas, S Murugesan, **M Beyeler** (2025). Chromatic encoding in the foveal bipolar layer: Implications for retinal prostheses. *Society for Neuroscience (SfN)* '25, San Diego, CA.
- A52 J Granley, LM Turkstra, G Pogoncheff, F Grani, L Soo, A Rodil, C Soto, TC Sprague, E Fernandez, **M Beyeler** (2025). Computational methods for extracting neural correlates of working memory and mental imagery from intracortical recordings in human visual cortex. *Vision Sciences Society (VSS)* '25, St. Pete's Beach, FL (**oral**)
- A50 L Gil Nadolskis, G Pogoncheff, J Granley, A Rodil, L Soo, LM Turkstra, TC Sprague, A Alfaro Saez, C Soto Sanchez, E Fernandez Jover, **M Beyeler** (2024). Sustained stimulus-selective multi-unit activity in human primary visual cortex. *VSS* '24, St. Pete's Beach, FL (**oral**)
- A45 A Xu, **M Beyeler** (2023). A biophysically detailed model of retinal degeneration. *COSYNE* '23, Montreal, Canada (poster, **A Xu: Travel Award**)
- A37 T Bhatia, Y Hou, J Granley, B Johnson, **M Beyeler** (2021). Nonlinear interactions with the retina shape the artificial vision generated by a bionic eye. *Society for Advancement of Chicanos/Hispanics and Native*

Americans in Science (SACNAS) National Diversity in STEM Conference (NDiSTEM) '21, online. (**T Bhatia: Undergraduate Poster Presentation Award**)

- A34 **M Beyeler**, GM Boynton, I Fine, A Rokem (2020). Interpretable machine-learning predictions of perceptual sensitivity for retinal prostheses. *ARVO '20*, Baltimore, MD. (**Abstract of Distinction, top 3 %**; canceled, COVID-19)
- A25 **M Beyeler**, El Yucel, A Rokem, GM Boynton, I Fine (2018). Optimizing stimulation protocols for prosthetic vision based on retinal anatomy. *COSYNE'18*, Breckenridge, CO. (oral)
- A19 GM Boynton, A Rokem, **M Beyeler**, J Dorn, NC Sinclair, MN Shivdasani, MA Petoe, R Hornig, I Fine (2017). Efficient and scalable measurements of sensitivity for high resolution electrode arrays. *The Eye & the Chip '17*, Detroit, MI. (poster, **Best Poster Award**)
- A16 **M Beyeler**, A Rokem, GM Boynton, I Fine (2017). Modeling the perceptual experience of retinal prosthesis patients. *VSS'17*, St. Pete's Beach, FL. (oral)
- A6 **M Beyeler**, M Richert, N Oros, N Dutt, JL Krichmar (2014). A cortical spiking neural network model for visually guided robot navigation. Neurobiologically Inspired Robotics workshop, *ICRA'14*, Hong Kong, China. (oral, **Best Student Talk Award**).

INVITED TALKS & SEMINARS

Scheduled

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| T48 Data-Driven AI for Neuroscience, <i>HHMI-Janelia Research Campus</i> | Nov 2026 |
| T47 Institute for Sensing and Embedded Network Systems Engineering (ISENSE),
<i>Florida Atlantic University</i> | Sep 2026 |

Selected Past Talks

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| T46 Laguna Blanca High School, <i>Santa Barbara, CA</i> | Feb 2026 |
| T45 Center for Responsible Machine Learning Agentic AI Summit, <i>UCSB</i> | Jan 2026 |
| T43 REI Talks, Brazilian Society of Low Vision, <i>YouTube (live)</i> | Sep 2025 |
| T42 World Blindness Summit, <i>Sao Paulo, Brazil</i> | Sep 2025 |
| T40 Computer Science Colloquium, <i>University of California, Riverside</i> | May 2025 |
| T39 Attention & Perception Seminar, <i>University of Illinois Urbana-Champaign</i> | Mar 2025 |
| T38 Scientist Society, <i>Fremont High School</i> | Mar 2025 |
| T37 Artificial Vision 2024: The International Symposium on Visual Prosthetics, <i>Aachen, Germany</i> | Dec 2024 |
| T35 Learning to see again: The role of perceptual learning & user engagement in sight restoration,
<i>VSS Symposium, St. Pete's Beach, FL</i> | May 2024 |
| T34 Neuralink Corporation, <i>Fremont, CA</i> | Jan 2024 |
| T33 Eye & Chip World Congress on Artificial Vision (plenary), <i>Detroit Institute of Ophthalmology</i> | Oct 2023 |
| T32 Cortical Prostheses Workshop: Interdisciplinary Research Towards Artificial Vision for the Blind,
<i>Hanse-Wissenschaftskolleg Institute for Advanced Study, Delmenhorst, Germany</i> | Sep 2023 |
| T31 Institute of Neuroinformatics, <i>ETH/University Zurich, Switzerland</i> | Aug 2023 |
| T30 Vision and Color Summer Data Blast, <i>Optica Society</i> | Aug 2023 |
| T29 60th Anniversary Conference, <i>Smith-Kettlewell Eye Research Institute, San Francisco, CA</i> | Aug 2023 |
| T29 Center for Visual Science, <i>University of Rochester</i> | Jul 2023 |
| T28 Hybrid Human-Machine Intelligence Summit, <i>DEVCOM Army Research Lab, Stanford, CA</i> | Jun 2023 |
| T27 Neuroscience + AI Seminar, <i>Carnegie Mellon University</i> | Apr 2023 |
| T26 20th Annual World Congress of the Society for Brain Mapping & Therapeutics (SBMT),
<i>Los Angeles, CA</i> | Feb 2023 |
| T25 SB Braille Institute, <i>Santa Barbara, CA</i> | Jan 2023 |
| T23 Optica Fall Vision Meeting, <i>Rochester, NY</i> | Oct 2022 |
| T22 San Marcos High School, <i>Goleta, CA</i> | Oct 2022 |

T21	NeuroVision Workshop, <i>CVPR '22, New Orleans, LA</i>	Jun 2022
T20	Translational Neuroengineering Technologies (TNT) Network, <i>Johns Hopkins University</i>	Apr 2022
T19	Tri-Service Research Laboratory (TSRL), Air Force, <i>JBSA-Fort Sam Houston, TX</i>	Mar 2022
T18	Universidad Miguel Hernandez, <i>Elche, Spain</i>	Feb 2022
T17	Claremont Colleges, <i>Claremont, CA</i>	Oct 2021
T16	Eye & Chip World Congress on Artificial Vision (plenary), <i>Detroit Institute of Ophthalmology</i>	Oct 2021
T14	14th Conference on Learning & Memory: Cellular and Systemic Views (canceled, COVID-19), <i>Leibniz Institut für Neurobiologie, Magdeburg, Germany</i>	Mar 2020
T13	Department of Cognitive Sciences, <i>University of California, Irvine, CA</i>	Apr 2019
T12	Department of Computer Science, <i>Duke University, Durham, NC</i>	Mar 2019
T11	Department of Computer Science, <i>University of California, Santa Barbara, CA</i>	Jan 2019
T10	Recent Advances in Neuroengineering Workshop, <i>COSYNE '18, Breckenridge, CO</i>	Mar 2018
T9	Center for Applied and Translational Sensory Science (CATSS), <i>University of Minnesota, Minneapolis, MN</i>	Feb 2018
T7	Cluster of Excellence in Cognitive Interaction Technology (CITEC), <i>Bielefeld University, Germany</i>	Aug 2017
T6	Center for Perceptual Systems, <i>University of Texas, Austin, TX</i>	Jul 2017
T4	Second Sight Medical Products Inc., <i>Sylmar, CA</i>	Nov 2016
T3	Department of Psychology, <i>University of Washington, Seattle, WA</i>	Dec 2015
T2	IBM Research, <i>San Jose, CA</i>	Aug 2015
T1	Qualcomm Technologies Incorporated, <i>San Diego, CA</i>	Nov 2014

TEACHING ACTIVITIES

Undergraduate Courses

UC4	CS-165A: Artificial Intelligence, <i>UCSB</i>	W'24
UC3	PSY-132: Visual Neuroscience, <i>UCSB</i>	S'23, S'25
UC2	CS/ECE-181: Introduction to Computer Vision, <i>UCSB</i>	W'21, F'22, F'24, W'26
UC1	PSY-130: Sensation & Perception · Vision, <i>UCSB</i>	F'20

Graduate Courses

GC3	CS/ECE-281B: Advanced Topics in Computer Vision, <i>UCSB</i>	W'23, W'26
GC2	PSY-221F: Computational Neuroscience	S'22, S'24, S'26
GC1	CS-291A: Bionic Vision, <i>UCSB</i>	W'20, F'21, W'25

Teaching Publications

10k copies sold

TP5	M Gevorgyan, A Mamikonyan, M Beyeler (2020). <i>OpenCV4 with Python Blueprints, Second Edition. Packt Publishing Ltd.</i> , Birmingham, UK, 366 pages, ISBN 978-178980181-1.
TP4	A Sharma, VR Shrimali, M Beyeler (2019). <i>Machine Learning for OpenCV 4, Second Edition. Packt Publishing Ltd.</i> , Birmingham, UK, 420 pages, ISBN 978-178953630-0.
TP3	M Beyeler (2017). <i>Machine Learning for OpenCV. Packt Publishing Ltd.</i> , Birmingham, UK, 382 pages, ISBN 978-178398028-4. Also available in Chinese, Japanese, Korean, and as a video course. [Code]
TP2	J Howse, P Joshi, M Beyeler (2016). <i>OpenCV: Computer Vision Projects with Python. Packt Publishing Ltd.</i> , Birmingham, UK, 558 pages, ISBN 978-178712549-0.
TP1	M Beyeler (2015). <i>OpenCV with Python Blueprints. Packt Publishing Ltd.</i> , Birmingham, UK, 230 pages, ISBN 978-178528269-0. [Code]

SCIENCE COMMUNICATION & PUBLIC OUTREACH

Public Lectures

PL4	Harold J. Plous Award Lecture, <i>UCSB</i>	2025
PL3	ARC Summer Faculty Series, <i>UCSB</i>	2025
PL2	UCSB Groundbreaking Research/Innovative Technology (GRIT), <i>UCSB</i>	2022
PL1	UCSB Open House (formerly 'Spring Insight'), virtual lecture, <i>UCSB</i>	2020

Selected Media Coverage

M15	Thrilling progress in brain-computer interfaces from UC labs, <i>UC News</i>	2025
M13	BionicVisionXR using Unity Sentis, Unite 2024, <i>Unity Technologies</i>	2024
M12	Towards a Smart Bionic Eye: YouTube, NNLM Discovery Podcast, <i>National Library of Medicine</i>	2024
M11	UCSB professor receives NIH Directors New Innovator Award, <i>Daily Nexus</i>	2023
M7	Are we witnessing the dawn of post-theory science?, <i>The Guardian</i>	2022
M6	Will it be possible to upload information to my brain?, <i>Gizmodo Asks</i>	2021
M4	Building the bionic eye... with car tech?, <i>PCMag</i>	2021
M1	Restoring vision with bionic eyes: no longer science fiction, <i>PCMag</i>	2019

Panels

PS4	NEI / RPB Mentored Career Awardee Symposium, <i>ARVO</i>	2026
PS3	Neuroscience Career Panel, <i>Academic Data Science Alliance (ADSA)</i>	2023
PS2	Demystifying the K99/R00 application at the National Eye Institute (NEI)	2021
PS1	An Evening with Neuroscience, <i>UW</i>	2019

Community Involvement & Public Outreach

CI8	Competition judge: UC LEADS	2026
CI7	Competition judge: Global Undergraduate Awards, <i>Dublin, Ireland</i>	2021 – 2024
CI6	Competition judge: SBHacks Hackathon, <i>UCSB</i>	2020 – 2021
CI5	Competition judge: US Congressional App Challenge, <i>Washington, DC</i>	2019 – 2020
CI4	Outreach & fundraising: Lighthouse Foundation for the Blind, <i>Seattle, WA</i>	2018
CI3	Neuronline community leader, <i>Society for Neuroscience (SfN)</i>	2016 – 2017
CI2	Student volunteer, <i>IEEE Robotics & Automation Society (RAS)</i>	2014 – 2016
CI1	Lab tour leader: Mathobotix "Bytes and Bots" K-12 Summer Camp, <i>UCI</i>	2013, 2014

PROFESSIONAL ASSOCIATIONS

- Member: *IEEE Engineering in Medicine & Biology Society (EMBS)* 2019 – present
- Member: *Association for Computing Machinery (ACM)* 2019 – present
- Member: *Association for Research in Vision & Ophthalmology (ARVO)* 2018 – present
- Member: *Vision Sciences Society (VSS)* 2017 – present
- Member: *Society for Neuroscience (SfN)* 2013 – present
- Neuronline Community Leader, 2016 – 2017

REJECTIONS & FAILURES

An attempt to normalize 'failure' in academia. Inspired by: Melanie Stefan (2010), A CV of Failures. *Nature* 468(467).
 Legend: T tenured, TT tenure track, PD postdoc, PhD grad

Academic Positions

Success rate, TT: 3 % (n=31), PD: 100 % (n=2), PhD: 50 % (n=2)

- Tenure-track positions (R1): 17 no answers, 12 rejections, 1 silent rejection after interview 2019
- Rockefeller University, Postdoctoral Position: accepted, offer declined 2016
- EPFL Neuroscience Graduate program: rejected 2013

Professional

Success rate, TT: 42.8 % (n=7)

- Harold J. Plous Memorial Award: not selected 2023
- MICCAI area chair: not selected 2021
- Next Generation Leaders Council at the Allen Institute for Brain Science: not selected 2020
- OCNS program committee: invited to apply 2019

Extramural Grants & Major Awards

Success rate, T: 25 % (n=4), TT: 40 % (n=15)

- NIH R01: not discussed 2026
- NIH R24: competitive, not discussed 2026
- Brain Research Foundation Seed Grant: selected at campus level but not awarded, role: PI 2026
- NIH R21: discussed but not awarded, role: PI 2025
- Moore Inventor Fellowship: not selected, role: PI 2024
- DARPA ICS: not awarded, role: co-PI 2023
- NIH R01: discussed but not awarded, role: co-PI 2023, 2024
- AFOSR Young Investigator Program: not invited for full proposal, role: PI 2023
- National Science Foundation (NSF) CAREER: not awarded, role: PI 2023 – 2025
- NSF AI Institutes: not awarded, role: co-PI 2023
- NSF Research Traineeship (NRT): not awarded, role: co-PI 2023, 2024
- Office of Naval Research (ONR) Special Notice: invited for full proposal, role: co-PI 2021
- SONY Focused Research Award: not awarded, role: co-PI 2021
- Chan Zuckerberg Institute (CZI) Essential Open Source Software: not awarded, role: PI 2020
- NSF NeuroNex: invited for full proposal, role: co-PI 2020
- ADSA seed grant: finalist, role: co-PI 2019
- Burroughs Wellcome Award at the Scientific Interface (CASI): invited for full proposal, role: PI 2018

Fellowships & Travel Awards

Success rate, TT: 42.9 % (n=7), PD: 100 % (n=4), PhD: 44 % (n=9)

- Sloan Research Fellowship: not awarded 2023 – 2025
- Microsoft Research Faculty Fellowship: not awarded 2021
- IJCNN Travel Award: not awarded 2015
- NVIDIA Graduate Fellowship: not awarded 2013 – 2015
- Microsoft Research Fellowship: not awarded 2013

Scientific Peer Review

- J26, *JoV*: rejected from a neuroscience journal 2026
- J24, *JNE*: rejected from two top-tier CS conferences 2025
- C24, *ICLR*: rejected from a top-tier CS conference 2025
- C23, *VRST*: rejected from an ACM conference 2025
- C20, *ISCA*: rejected from a top-tier CS conference 2025
- J21, *Scientific Reports*: rejected from two top-tier HCI conferences and a clinical journal 2025
- J20, *Translational Vision Science & Technology*: rejected from a top-tier HCI conference 2024

- J18, *Journal of Neural Engineering*: rejected from an IEEE journal 2023
- J16, *Frontiers in Neuroscience*: rejected from 2 high-impact neuroscience journals 2023
- J15, *Journal of Vision*: rejected from 2 top-tier HCI conferences and 1 translational journal 2023
- C12, *Augmented Humans (AHs)*: rejected from a top-tier HCI conference 2022
- W2, *MICCAI-W*: rejected from main conference track 2021
- J8, *Scientific Reports*: desk-rejected from 5 high-impact neuroscience journals 2018
- J7, *Frontiers in Neuroscience*: desk-rejected from 1 high-impact neuroscience journal 2018
- J6, *PLOS Computational Biology*: desk-rejected from 3 high-impact neuroscience journals 2017
- COSYNE abstract: rejected 2015, 2018

Workshops

Success rate, TT: 60 % ($n=5$), PD: 50 % ($n=2$)

- VSS workshop proposal: rejected 2025
- NeurIPS workshop proposal: rejected 2021
- VSS workshop proposal: rejected 2019